Journal of Business Venturing Insights 14 (2020) e00188

Contents lists available at ScienceDirect

Journal of Business Venturing Insights

journal homepage: www.elsevier.com/locate/jbvi

The roles of thought and affect on entrepreneurship – A new hope☆

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ARTICLE INFO

Keywords:
Affect
Hope
Entrepreneurial process
Cognition
Neuroscience

ABSTRACT

We challenge the current, psychology-based model of affect's influences on the entrepreneurial process (and some of its past empirical support). We review the current model and identify its limitations. We analyze the relevance of 'hope' as a cognitive capability underlying the entrepreneurial process and assess why it has yet to be leveraged in our field. We then use hope and the recent studies on emotional control to argue a new, and more prescriptive model of the role of affect in entrepreneurial cognition and action.

We question the existing syllogism: Emotions influence thought. Thoughts influence entrepreneurial decisions. And, thus, emotions are important to the entrepreneurial process (e.g., Baron, 2008; Baron and Tang, 2011; Foo, 2011). Specifically, we question the implied assumption that emotional influences are uncontrollable. Instead, we argue that emotional control is practically feasible. As such, we suggest that the reports of the impact of *affect* on the entrepreneurial process have been greatly exaggerated. Rather than being a significant *exogenous* force, affect should be considered as just another factor that *can* influence decision-making, although in relatively complex ways. Our contribution here is to challenge the current model (and some past empirical support) of affect's influences on the entrepreneurial process by providing a better and more prescriptive model, one based on *actual hope* and recent studies.

1. Affective (but not necessarily effective) entrepreneurship

We follow Baron's (2008) seminal work in defining the relevant terms here. *Affect* represents the emotions, moods, and passions of the entrepreneur – it is the realized subjective feelings of (dis)pleasure attached to an object (e.g., experience; physical thing), where such feelings that can differ in length, intensity, antecedent, and so on. The *entrepreneurial process* consists of creatively identifying an opportunity and acting to exploit it, all under uncertainty (where the opportunity can be identified *ex post* as a market imperfection believed *ex ante* to indicate that unrealizable value existed from which some party can profit). That process is often characterized by multiple stages, non-routineness, and imperfection. Because such characterizations are associated with higher levels of emotion (e.g., where the personal stakes and pressures are greater – Cardon et al., 2012), the entrepreneurial process requires attention be paid to affect.

Baron (2008) highlights the many *potential* impacts of affect in the entrepreneurial process (e.g., on creativity, persuasion, decision-making, judgment, and inter-personal relationships) with a solid framework based on select empirical research. He cites neuroscience-based studies that reveal how affect and cognition interact at very basic levels of neural functioning during problem-solving and decision-making salient to the entrepreneurial process.[1]

☆ No human subjects or animal participants were involved in this research.

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[1] The relevant neuroscience research is based partly on *f*MRI studies where human subjects play economic games. These reveal the brain's emotion centers lighting up more when the subjects' decisions went against their own economically-rational interests (suggesting the power of affect over cognition).

https://doi.org/10.1016/j.jbvi.2020.e00188
Received 30 April 2020; Received in revised form 10 July 2020; Accepted 13 July 2020
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He also cites our field's empirical evidence where affect has been shown to influence important aspects of the entrepreneurial process, like creativity (e.g., where positive affect correlates with increased creative cognition and alertness).

A recurring theme is that affect's influences (on outcomes) are complex. The complexity arises from two main characteristics of the potential relationships. First, the influences can take both a direct and an indirect form. Affect can directly influence outcomes (e.g., as a separate input that signifies a non-cognitive judgment about a choice). And, affect can indirectly influence outcomes by impacting cognitive functioning (e.g., by priming memories, by providing cues for inferred reactions, and so on). Second, the influences are not uniform. The influence can not only be of a positive or negative type (e.g., improving or impairing cognitive functioning; increasing or decreasing risk perception) but also that type of affect can have either impact (e.g., positive affect can both help and hinder the process outcomes). That complexity does mean that there is no easy approach to emotional control (e.g., suppression). But it does *not* mean, as Baron (2008) suggests, that all studies and theorizing needs to include affect when investigating the entrepreneurial process (or entrepreneurial cognition). Indeed, many factors have complex effects on processes, but if they are controllable, then they just become part of the internal workings assumed in an efficient and effective approach to that process (rather than a special set of exogenous issues that require independent analysis).

More recent work builds on Baron's (2008) ideas, such as Welpe et al.'s (2012) research on emotion and entrepreneurial exploitation. They formalize the direct and indirect effects of emotion. *Affect-as-information* captures the direct judgment impact of emotion on outcome through appraisal (e.g., where anger [non-cognitively] reduces the risk perception of a choice). *Affective-processing* captures the indirect impact on cognitive evaluations (e.g., with emotion influencing cognitive content). In their thought-experiments on student subjects, they found that although cognition mattered by far the most, important outcomes were influenced by affect (directly, and indirectly through an interdependency with cognition). Notably, such research *where affect is induced* proves the point that affect *is* controllable in its complex effects on the (simulated) entrepreneurial process. (That begs the question of *why can't the subjects be taught that same control for their self-benefit?*) And, studies have begun to focus on managing affect, under labels like 'controlling the heart', and 'emotional labor', with limited results. "*We do not have much evidence to date on how emotional labor might shape the entrepreneurial process*" (Cardon et al., 2012: p4).[2]

2. Hope (theory) as an (ironically) wasted potential path in entrepreneurship research

In the recent psychology literature, *hope* is defined as a cognitive capability (Snyder, 2002) and *not* as an emotion (Shorey et al., 2002). Unfortunately, in the entrepreneurship literature, it has been mischaracterized as an emotion or ignored altogether (as it was in Baron's (2008) framework). This not only raises concerns over the legitimacy of that entrepreneurial emotion stream but it is wasteful; it is wasteful because as a cognitive capability the conceptualization of hope aligns so closely with the core capability to complete the entrepreneurial process.

Snyder's (2002) seminal work on hope (and its theory) provides a solid description of the construct (along with several readily-implementable and verified measures). "*Hope is defined as the perceived capability to derive pathways to desired goals, and motivate oneself via agency thinking to use those pathways*" (Snyder, 2002, p249). Hope, like entrepreneurial activity, is centered on goals, pathways and agency. Hope, like entrepreneurship, centers on goals characterized by some level of perceived or self-added uncertainty (e.g., stretch goals). Hope, like entrepreneurship, regards goals as unanswered calls that initially lack the requisite means to reach them. And, hope is aligned with the kind of foresight that improves the entrepreneurial process (e.g., by allowing the entrepreneur to see the implications of her actions, to detect and avoid problems, to see the effects of possible future events, and to envision desired futures – Morrow, 2006). In fact, it appears that hope *is* entrepreneurship when the focal goal is explicitly defined by new value creation (although even that specification can appeared to be implied, at least for the focal agent).[3]

We see hope as one capability for effectively controlling emotions in the entrepreneurial process. We consider hope as a primary pre-cognitive method (Morrow, 2006) for increasing the mindfulness in order to be successful in the process. We recommend equipping entrepreneurs with high hope levels in order to provide a means for dealing with affect with the energy, creativity, foresight, and focus on the implementation activities required. Give entrepreneurs the tools to generate their pathways to their goals by breaking those into more proximate and reachable sub-goals that effectively induce and exploit the affect they need to improve their decisions and outcomes.

Hope is a learned thinking pattern, and a complex one (just like entrepreneurial cognition). Hope requires not only the consideration of one's own goals but also the consideration of the desires and capabilities of others. (This makes managerial choices, like delegation, supportable as part of hope's pathways.) Hope involves two major types of goals – there are positive goals (e.g., to newly reach, sustain, or increase some beneficial target outcome) and negative goals (e.g., to deter finally, or to delay, some harmful outcome). (Such positive

[2] Older constructs, like 'emotional intelligence' [EI], touched on this previously, in terms of having one component of its lengthy definition being 'the use of emotions to achieve goals' (Salovey and Mayer, 1990). However, the emotional control measured in *EI* so far has been more about suppression of negative emotions than about the more subtle control of impacts in a complex process.

[3] Snyder's (2002) definition of hope even describes several real observed behaviors of entrepreneurs. For example, high-hope agents are persistent yet flexible, understand their goals and are invigorated by them, just like successful entrepreneurs. Further, high-hope agents will re-goal (i.e., pivot) when they are stumped (doing so to find proximate and reachable new goals), just like many entrepreneurs have described in their own journeys. Hope theory emerged in the broader scope of psychological research in the 1980s, in a first-person sense, when subjects were asked about what was driving their behaviors, in this case as a positive side to explaining failures (when reaching for goals). Such research considers not only affect and cognition but also their interdependencies.

goals are normally associated with regular entrepreneurship [e.g., in bring a new product to market] while the negative goals appear more associated with social entrepreneurial activities [e.g., to delay the harmful effects of global warming].) Hope, like the entrepreneurial process, involves dynamics and learning (e.g., where pathways and agency feed each other). 'To hope' is *not* to be naïve; indeed, barriers to goals (characterized as problems) are expected, just as in most entrepreneurial pursuits. And hope, like (entrepreneurship-related) affect, also has a neurobiological basis – given it has been proposed in neuroscience that the very purpose of the human brain is to find pathways to goals (Craig, 1943).

Hope theory originated from, and is oriented towards, a first-person research approach rather than the third-person approaches and causal models and the subsequently limited methodologies common to the past psychology-based work on entrepreneurship (Dimov et al., 2020). Hope-as-cognition is a fresh divergence from the mainstream psychological research that too often has taken the entrepreneur 'as given', rather than as one of a subset of individuals with potentially entrepreneurial capabilities (like hope) who chose to venture when others did not (perhaps due to contextual factors) (Ramoglou et al., 2020). Such divergent perspectives help increase our understanding of the agency, capabilities and personal circumstances involved in the entrepreneurial process.

Several possible reasons for why our field has *not* exploited Synder's (2002) characterization of hope exist. First, it runs counter to hyped logics of the time that assumed expert entrepreneurs' actions were not goal-directed but instead means-directed (e.g., Sarasvathy, 2001). Second, it runs counter to some definitions of the related term 'hopeful', which is a passive and excusing approach or state feeling. Third, it relegates emotion to *reactive* role (very explicitly), which undercuts the impact of the entrepreneurial emotion stream. In hope theory, emotions follow cognitions, and so are secondary to cognitive appraisals: "… *the protagonist's emotions do* ***not*** *make his or her decisions – he or she does, through conscious choice*" (Shorey et al., 2002: p328). Further, affect-focused empirical work is questioned: "… *beyond the lack of reliability within and across people in identifying the meanings of specific emotions, we also do not see the usefulness of this process"* [of identifying emotions in the hope-related action process] (Shorey et al., 2002: p327). And, even when strong emotions follow an action's outcomes, in hope theory, they are minimized. For example, a failure to achieve a goal informs adaptive (high-hope) agents merely that they had not chosen the right paths to attain their goals, rather than reflecting something innately bad about themselves as a negative affect would infer.

But, perhaps the main reason for the wasting of the hope concept lies in its unfortunate misrepresentation in the entrepreneurship empirical literature. We consider two examples: Huang et al.'s (2019) recent piece in *SMJ* and Foo's (2011) less recent work in *ETP*, both of which used young students in simulated decision-making. Huang et al. (2019) focused on escalating commitment (in the face of failure), analyzing the impacts of emotional dynamics in the team of co-founders on the entrepreneurial decision of whether or not to shut down their new (simulated) venture. They argued that hope – characterized as an emotion – would have both direct and indirect impacts on the decisions. They used one item to assess hope (i.e., *My team feels hopeful that we will succeed in the game*). But that was problematic in several ways: that item is **not** one of the actual hope state scale items in the source they quoted (which was actually Synder, 2002); and, despite hope explicitly defined as **not** being an emotion in the citation, they chose to mischaracterize it as such. Regardless, their hope measure had a significant effect (directly) on escalating commitment (and trumped the effects of group fear). Given the problems noted, it is difficult to know what was actually measured (other than to note it was *not* the hope concept that was cited), and whether it actually had a beneficial influence on (entrepreneurial) performance because that was actually *not* reported (i.e., as the financial performance of the simulated ventures was not used as a dependent variable, even though it was clearly available). So, altogether, this piece was not a high point of work in our field for increasing the understanding of hope-as-cognition (or hope theory) in entrepreneurship.

Foo's (2011) study also involved a fictitious venture where they induced, rather than contemporaneously measured, affect in student participants. Questions were used to induce one of four emotions (i.e., anger, happiness, hope and fear), where the definition of the emotion was left entirely to the participant. The impact of the inducted emotion was then tested on the participant's risk perception about a standard hypothetical venture. This study characterized hope as indicating the most risk perceived (i.e., hope signified uncertainty and a lack of control). As such, this is another example of recent work in our field that does *not* adhere to an established measure (and definition) of an existing construct (apparently opting for no actual definition), while expressing confidence in reaching conclusions about it (without even citing its literature).[4]

3. Relegating affect to another cognitive tool in a hope-ful best case

Consider Fig. 1 as depicting a layered image of the ideal versus the real entrepreneurial process. Ideally, outcomes (as entrepreneurial choices and actions) are solely products of rational thinking based on good information (as indicated in the figure's inner relationship). But, this scenario is *un*likely in any actual entrepreneurial context. In reality, because decision-makers are human, and humans experience emotions, and emotions can influence outcomes directly and indirectly (by impacting cognitive processing), the link between thinking and outcomes is *not* ideal (e.g., Baron, 2008). (That is depicted as the first outer relationship.) But, given the array of research (across many fields) on emotional induction and its effects on decision-making, we should hypothetically be able to manage, and even exploit, affect. We should be able to effectively endogenize its influence and make it just one more controllable factor (like the temperature in an office), even for the messy process that is entrepreneurship and the complex factor that is affect. (This is depicted in the outer relationship.)

In effect, we agree with the story told by Baron (2008) and others regarding the *potential* impacts of affect on the entrepreneurial

[4] Although the study had very questionable conclusions about hope, it was hard not to agree with its stated implications about emotions – e.g., that emotional skills should be taught in entrepreneurship, and that decisions should be delayed when strong emotions are at play.

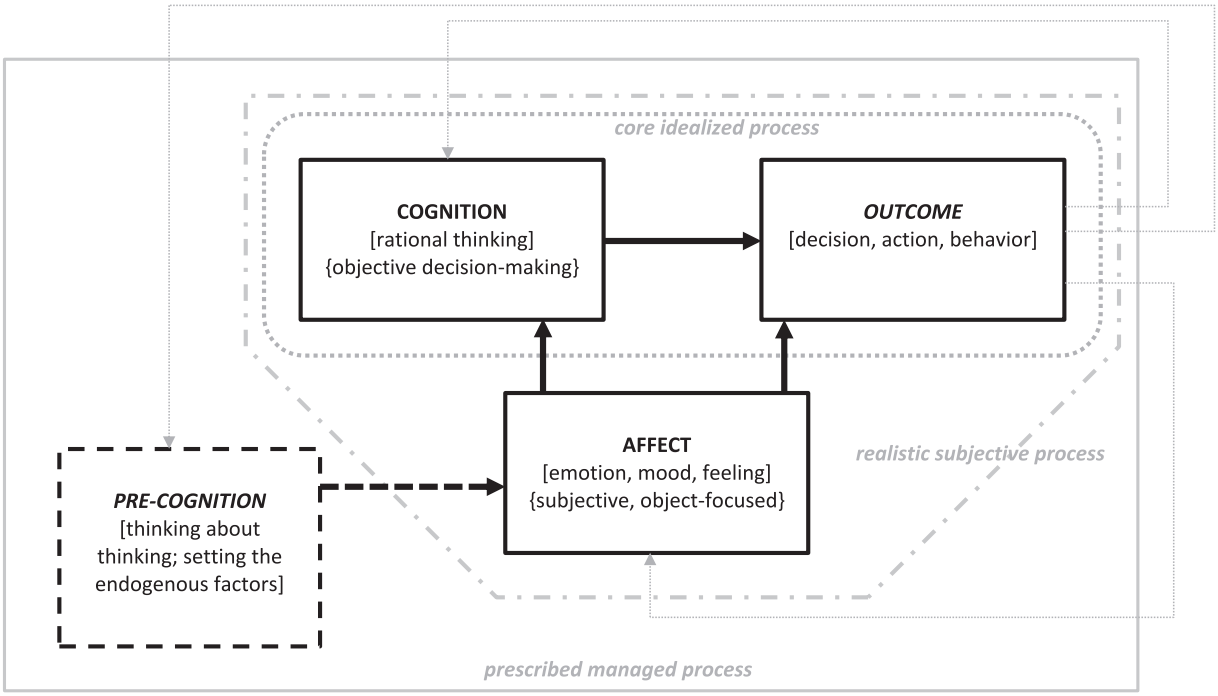

Fig. 1. Layers of thought, affect and outcome for the entrepreneurial context. NOTES: (1) All processes are imperfect in the entrepreneurial context, as: decisions are non-routine; manager cognitive resources are limited; information is noisy and constrained; and, so on. (2) Four-part dotted-line arrows indicate feedback in the process for the entrepreneur/manager to learn from.

process, but we strongly disagree that is the full story or even the most important part of the story. And, we see that realization in some of the relevant (but not yet entrepreneurship) literatures. For example, Seo and Barrett's (2007) *AMJ* piece exploring the link between affective experience and decision-making performance found that the individuals who displayed a higher control of their feelings had better performance (in simulated investments). That emotional control was characterized as a learned cognitive skill that included recognition, differentiation, and regulation of affect. They determined that "… *whether affective feelings actually hurt or help decision making can be largely determined by how individuals experience and handle those feelings in more or less functional or dysfunctional ways* …" (p925). One trick they identified involved removing the emotional bias in a decision simply by attributing feelings to the correct object.[5] Such adaptive strategies provide affective cognitive control capacity (e.g., Aldao et al., 2010; Schweizer et al., 2013).

Another example is Clore and Huntsinger's (2009) work on affect and its attributed objects. "*The power and impact of any particular emotional moment would lie, then, in what the emotional reaction seems to be about. What specifically has the emotion marked as good or bad, what is at stake, and what new goals does it entail? In this view, emotions differ from each other because they have different kinds of objects or signify different kinds of situations*" (p48). They find that emotion is *not* a cause of an outcome but, instead, it *is* the outcome. In that story, the reaction to a poor outcome is the setting of new goals (perhaps as depicted in the feedback loops in our figure). In their story, emotions are multi-faceted representations of psychologically significant situations rather than the *de facto* precursors to thought; affect represents the memories of responses to real-world obstacles and opportunities. As such, affect can be manipulated to improve cognition (e.g., by framing the object of the emotional power to produce the desired influence directly and indirectly on outcomes). This entails (pre-)thinking before any emoting that precedes thinking (in order to break any chicken-egg ordering issues).

We proposed that affect is just another factor that should be controlled by cognition in an efficient overall process. Now, we understand that this thought might cause negative feelings for those involved in entrepreneurial emotion research, but we are hopeful that they will manage that impact effectively based on their own abilities to manipulate-through-induction the passions of others (e.g., by generating a misattribution of feelings to targeted objects). To bolster our proposition we highlighted the use of hope as one possible capability for managing the affects that could influence entrepreneurial decisions. And, we invite future research to explore this proposition further, for example, through experiments and case studies where hope (or another capability) is trained prior to the influence of affect (natural or induced) to better determine how well and under what conditions emotions can be managed and exploited

[5] They also found both the direct and indirect impacts of affect, largely in agreement with other studies like Clore and Huntsinger (2007): Directly, affect assigns value to whatever seems to be causing it, and provides judgments over prioritization and selection; used well, it offers a fast way to filter down options. Indirectly, affect influences the content and style of cognition through: memory priming; motivation; attention; access to prominent responses; false memories; forgetting; invocation of distinguishable frames of mind; and, relational (versus item-specific) focus (e.g., pleasant affect aids creativity by priming a broader perspective).

in the entrepreneurial process. We believe that is a better path forward – one based on a story with hope, and without all the unnecessary emotional baggage.

Credit author statement

This is the Author statement file outlining their individual contributions to the paper using the relevant CRediT roles: Conceptualization; Data curation; Formal analysis; Funding acquisition; Investigation; Methodology; Project administration; Resources; Software; Supervision; Validation; Visualization; Roles/Writing - original draft; Writing - review & editing. All roles were done by the sole author for this paper, Richard J Arend.

Funding

This study was not funded.

Declaration of competing interest

There are no conflicts of interest involved with this paper.

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